

LEARNING MODULE

4

Probability Trees and Conditional Expectations

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LEARNING OUTCOMES

<i>Mastery</i>	<i>The candidate should be able to:</i>
☐	calculate expected values, variances, and standard deviations and demonstrate their application to investment problems
☐	formulate an investment problem as a probability tree and explain the use of conditional expectations in investment application
☐	calculate and interpret an updated probability in an investment setting using Bayes' formula

INTRODUCTION

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Investment decisions are made under uncertainty about the future direction of the economy, issuers, companies, and prices. This learning module presents probability tools that address many real-world problems involving uncertainty and applies to a variety of investment management applications.

Lesson 1 introduces the calculation of the expected value, variance, and standard deviation for a random variable. These are essential quantitative concepts in investment management. Lesson 2 introduces probability trees that help in visualizing the conditional expectations and the total probabilities for expected value.

When making investment decisions, analysts often rely on perspectives, which may be influenced by subsequent observations. Lesson 3 introduces Bayes' formula, a rational method to adjust probabilities with the arrival of new information. This method has wide business and investment applications.